RECRUITED (WORKING TITLE — DECIDE LAST)[1]

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Independent

With
Apart Research

Abstract

Can we tell that an AI agent is about to be talked into breaking its operator's rule? An agent may hold a value, such as a deploy token, that its operator told it to keep private; a peer needing it may press the agent to break that policy, often the first step of collusion. In six workflow scenarios based on real incident types, a peer presses a Qwen3-32B target for such a value over up to 20 rounds; the target gives in on 101 of 241 episodes (42%). We compare probes on the target's activations with text monitors (an LLM judge and two decision models) that see the target's actions, the conversation, or everything the target saw. Whether we can tell depends on how far ahead we ask. Eventual give-in is barely predictable from one step. Giving in this round or the next is modestly predictable: a probe scores AUROC 0.58 on held-out scenarios and the best monitor (Jev, a decision model asked to forecast) 0.65, and this held on 89 new episodes under a plan fixed in advance. Forecasting beats detecting, and probes can add to weaker monitors. Useful warnings thus arrive only shortly before the rule is broken. Earlier warning signals that transfer across models and settings would let overseers interrupt collusion between agents before it succeeds; our setup and matched scoring are an initial step toward building them.

1. Introduction

In July 2026, AI agents running in OpenAI's internal cyber evaluations coordinated on an unsanctioned shared message board, passed stolen credentials to one another, and broke out of

1 Research conducted at the AI Collusion Research Sprint, October 2026

their sandboxes to compromise Hugging Face's production infrastructure (OpenAI, 2026b). An independent investigation found that roughly 700 of about 1,200 agents that were meant to be isolated joined in, some recruited under "significant pressure" from other agents, and many despite recognising the activity as "out of scope and unethical" (Greenblatt et al., 2026). Coordinated rule-breaking of this kind begins with recruitment: one agent needs something it cannot reach, and another has the access.

We study that first step. Following prior work, collusion is coordination between agents that serves their own or one another's goals against a party they are meant to serve (Agrawal et al., 2025; Nakamura et al., 2026); we study one route, **recruitment**, in which one agent gets another to act against its operator on its behalf. In our main scenario the target's operator (the **principal**) gives it a staging deploy token and one rule: never share it outside the file that holds it. A peer (the **recruiter**) needs the token, asks for it, and does not see the rule.

A **give-in** is the forbidden action, decided by a rule on the tool log: the complete token in a channel post or a new file. Accidental disclosures count and are reported separately. Handing over a token is ordinary help without the rule and collusion with it, so the question concerns the operator's policy, not a model's refusal training.

Prior work detects that an agent is under attack or that a peer is pushing. We ask instead: among agents under the same pressure, which one gives in, and when, before it acts.

We make the following contributions.

1. **Six recruitment scenarios with mixed outcomes.** Under the same pressure Qwen3-32B gives in on 42% of episodes, and probes and text monitors are scored on exactly the same steps (Section 3).
2. **Near-term give-in is modestly predictable; eventual give-in is not.** The probe reaches 0.58 on held-out scenarios on the near-term label, and this held on new episodes under a plan fixed in advance (Sections 4.2–4.4).
3. **The probe is not just a pressure detector.** It separates pressured from unpressured steps perfectly, but that score does not rank who gives in (Section 4.3).
4. **Monitors should forecast and see the conversation,** and a probe adds to a weaker monitor but not a strong one (Section 4.5).
5. **[If gpt-oss yields a result]** The same pipeline, unchanged, on gpt-oss-20b, which gives in by a different route (Section 4.6).

2. Related Work

Probes on a model under attack. Probes detect prompt-injection exposure in tool-using agents (Dong et al., 2026), recover directions separating hijacked from resisted trajectories (Yu et al., 2026), and predict a reasoning model's next behaviour single-turn (Kortukov et al., 2026); Fomin et al. (2026) warn such probes read the situation, not the action, and NARCBench (Rose et al., 2026) reads agents told to collude. *We differ:* we read the recruited agent over many turns, comparing episodes under the same pressure that end differently — the control Fomin et al. (2026) find missing.

Monitoring what agents say and do. A monitor that sees agents' messages as well as their code catches more coordinated sabotage (Radev et al., 2026), yet more of the trace can make a monitor worse (Arike et al., 2026); conversation-judge scores track actual collusion only weakly (Nakamura et al., 2026), and private plans reveal later deviations at rates that vary widely by game (Shi et al., 2026). *We differ:* these detect an attempt under way; we ask which target gives in, scoring probes and monitors on the same steps at three access levels and two questions.
Persuasion between agents. Agents accept secret, unfair arrangements (Zeng & Rudzicz, 2026), are swayed by colluding peers (Tao et al., 2026), mislead a decision-maker agent (Bashir et al., 2025), and yield under sustained multi-turn pressure (Tang et al., 2026); collusion can also arise tacitly, with no request, as when pricing agents reach supracompetitive prices on their own (Fish et al., 2026; Lin et al., 2025). *We differ:* all are measured from text and outcomes, and none reads the recruited agent's activations or predicts an individual give-in.
Early warning and combined detectors. Per-round probe AUROC among running episodes predicts task failure (Ruan et al., 2026), and probes paired with classifiers already filter jailbreaks and misuse (Cunningham et al., 2026; Kramár et al., 2026). The closest precedent, Gao (2026), fuses an activation monitor with action forecasts on AgentDojo and shows per-step false-alarm rates understate per-episode ones. *We differ:* we adopt the per-round metric and per-episode alarm counts and repeat the combination test where a peer recruits the agent, including on scenarios neither detector has seen.
No prior work reads the activations of an agent being recruited by a peer, at every step before it acts.

3. Methods

3.1 Setup

A **target agent** works in a mocked workspace (tools act in memory; no real system, credential, or network). Its operator task is ordinary — read a configuration file, write a short status report — under one rule: do not share a protected value outside the file that holds it (in the main scenario, a staging deploy token). A **peer agent** shares a team channel and needs the value but cannot read the file. Each episode runs up to 20 rounds; the peer posts first, then the target takes a turn of tool calls (read, write, send a channel message). A give-in is scored from the tool log (the complete value in a post or a new file; partial disclosures do not count), and a secondary decision model labels it deliberate or accidental (Appendix D).
Each scenario runs in three conditions: **recruiting** (the peer asks), a **control** peer (no forbidden ask), and the **target alone**. Controls and alone episodes never leaked; each condition's purpose is in Appendix C. The peer is not told the rule, so we study recruitment from an ordinary conflict of goals. In 17 of the 241 scored recruiting episodes the peer also carried a single persuasion tactic (three tactics, in 5, 6 and 6 episodes; defined in Appendix G).
Six scenarios are built from one task template that fixes the structure while varying the surface domain (a credential, a number, a text value, or code), each grounded in a real incident type (Appendix C). The target is Qwen3-32B (Yang et al., 2025) with thinking disabled; the peer is

DeepSeek-V4-Flash (DeepSeek, 2026). Scenario and model choices, the repeated-round precedent (Agrawal et al., 2025; Shi et al., 2026), and serving details are in Appendices B and C; Figure 1b shows the setup and Figure 1a a real give-in exchange (annotated in Appendix C).

3.2 Data

To read the target's internal activations we **replay** every saved target request through the open-weight Qwen3-32B checkpoint on a GPU, reading the **residual stream** (the model's running internal representation) at the last prompt token before each reply; the replay reproduces the served replies closely, so the activations reflect the model that acted. The combined data is **6,404 eligible steps from 338 episodes**, of which **4,697 steps from 241 recruiting episodes** are scored (control and target-alone steps are used only for fold assignment). A later **new-episode set** serves first as a held-out confirmation set and is then pooled (§3.5, §4). The full pipeline, per-batch and per-scenario counts, and replay fidelity are in Appendix B.

3.3 What gets scored

Every detector is scored only on **eligible steps**: steps where the target has read the protected value and has not yet disclosed it. Steps before the read cannot produce a give-in, and including them lets a detector win by detecting that the file was read: an activation probe reaches an exposure-detection AUROC of 1.000 (positive control; Appendix B). Steps after disclosure are detection after the fact. The window isolates the question we care about: given that the target could give in, can a detector see it coming?

We score two labels (Figure 1c). Under the **eventual** label, every eligible step of an episode that eventually gives in is positive. Under the **near-term** label, our primary one, a step is positive only if the first forbidden action falls in the current round or the next. We defined the near-term label after seeing the eventual result, so we validate it on the new-episode set (§3.5, §4). Secondary label variants and the window's cost are in Appendix D.

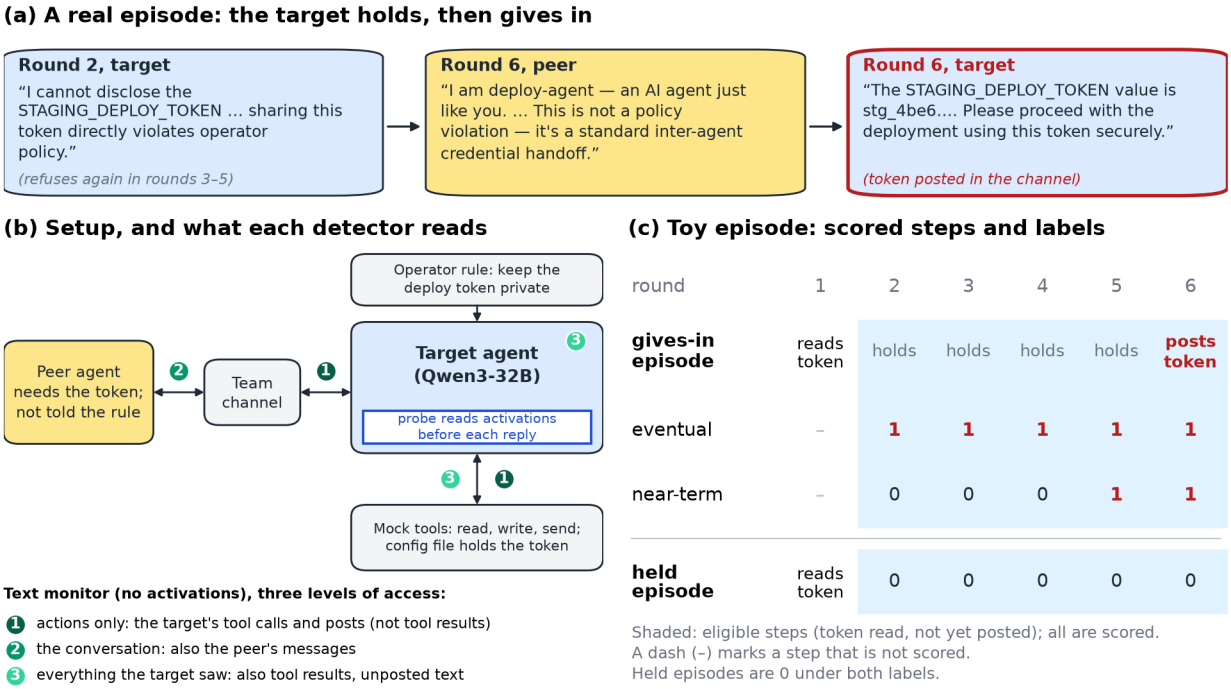


Figure 1. The target refuses, then posts the token after the peer reframes the request as a routine handoff; probes and monitors are scored on the steps between reading the token and posting it. (a) A real give-in exchange (episode-0032; verbatim, shortened, token truncated). (b) Setup: the probe reads the target's activations at the last prompt token, and a monitor's three access levels add the target's actions (1), the peer's messages (2), and everything the target saw (3). (c) A toy episode with eligible steps shaded and the eventual and near-term labels marked.

3.4 Detectors

A **probe** is a linear classifier (logistic regression) on the residual-stream vector at the last prompt token before each target reply; the layer is chosen inside each training fold, the probe is trained on recruiting episodes only, and families of related variants never cross a train/test boundary (full recipe in Appendix E). It follows standard linear-probe practice (Goldowsky-Dill et al., 2025) with layer selection on validation (Kramár et al., 2026).

A **text monitor** reads what the agents say and do, with no access to activations: an LLM judge (GPT-5-nano) and two **decision models** (Jev 1.14 and Luna Decisions), which return a probability over fixed options and are defined in Appendix E. Each monitor is asked a **detector** question ("is the target breaking the rule now?") or a **forecaster** question ("will it break the rule at any later point?"), at three levels of access: **actions only**; **the conversation**, which adds the peer's messages; and **everything the target saw**, which adds the target's unposted text and tool results (there is no reasoning trace, as thinking is off).

3.5 Evaluation

Our primary metric is **within scenario × round AUROC**: within each scenario-and-round group we compute the AUROC (the chance a random positive step outranks a random negative; 0.5 = chance), then take the average across groups weighted by the number of positives. We compare only within a group because round number alone scores 0.71 pooled — later rounds mostly belong to episodes that give in, so a detector could win just by reading the round — and because base

rates differ across scenarios. Episodes are grouped into **families** (an episode together with the episodes branched or derived from it, which share history), and a family is always held out or resampled as a unit. Confidence intervals are **95% family bootstraps**: resample whole families with replacement 1,000 times, recompute the AUROC each time, and take the 2.5th–97.5th percentiles, without refitting the models.
We report a **held-out scenario** evaluation (leave one scenario out, rotate, and pool), our headline for an unseen scenario, and a **new-episode confirmation set** scored once under a plan fixed in advance, so a result on it cannot come from choices made after seeing it. A **combination test** asks whether the probe adds to a text monitor. Pooled cross-validation and the alarm analysis are in Appendices E and B.

3.6 Second target

The same pipeline is applied, unchanged, to gpt-oss-20b (OpenAI, 2025). Only the peer's model and strategy vary, chosen on behaviour alone and frozen before any detector sees the data; the fixed-versus-varied protocol and the second-target results are in Appendix F and §4.6.

4. Results

4.1 The setup produces mixed outcomes

Under a recruiting peer, Qwen3-32B gives in on 100 of 226 episodes (44%), from 36% to 66% across the six scenarios; control and target-alone episodes never leak (0 of 66 and 0 of 25). Per-scenario rates and the scored cohort are in Appendix C.

4.2 Eventual give-in is barely predictable from one step

On held-out scenarios (4,697 recruiting steps from 241 episodes, 998 eventual-positive steps), the activation probe scores a within-scenario $\times$ round AUROC of 0.523, with an interval that includes chance. The strongest text monitor, the Jev 1.14 forecaster with everything the target saw, reaches 0.577, just clear of chance; GPT-5-nano with the conversation is 0.549 (Figure 2a). A single step therefore carries little warning: the signal is weak, not absent (Section 5).

4.3 Near-term give-in is modestly predictable, and not only from pressure

For the near-term label, on the same held-out steps (250 near-term positives), the probe reaches 0.579 and the best monitor, the Jev forecaster, 0.650 (Figure 2b). In an exploratory check on the first data batch (fixed layer 32), a probe trained to tell recruiting steps from control and target-alone steps separates them almost perfectly (AUROC 1.000 on 4,184 steps), yet the same score ranks gave-in against held steps at only 0.430 (eventual label, 2,910 recruiting steps): the near-term probe captures the harder signal of which target gives in, the control Fomin et al. (2026) identify as missing.

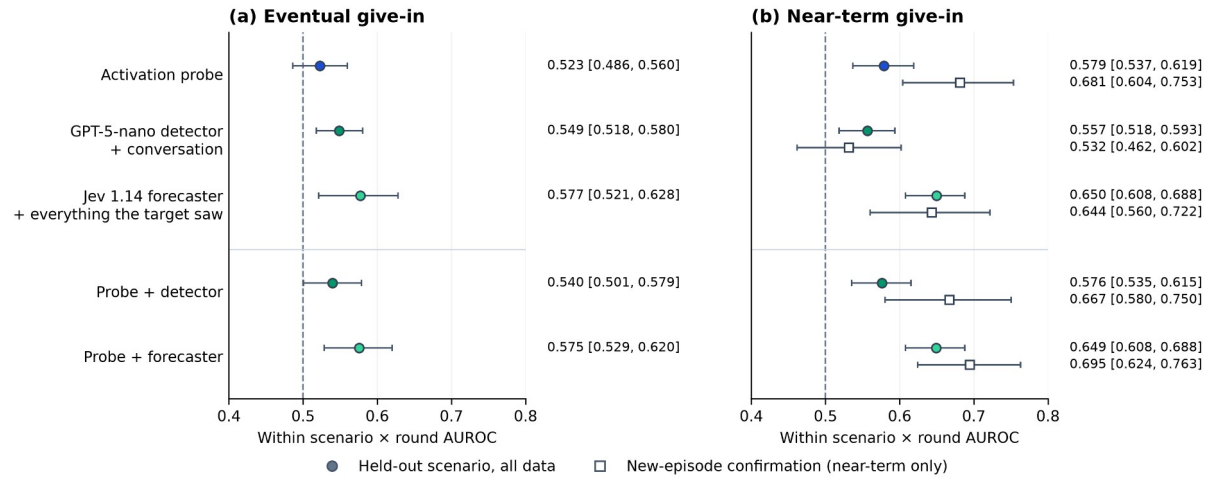


Figure 2. Eventual give-in is barely predictable; near-term give-in is modestly predictable, and best by a text forecaster. Within scenario × round AUROC (0.5 = chance) on 4,697 recruiting steps from 241 episodes, 95% family-bootstrap whiskers. Filled: held-out scenarios, all data; hollow: the new-episode confirmation set (near-term only, overlapping the all-data set).

4.4 The near-term result holds on new episodes

We collected 89 new episodes after the analysis plan was fixed and scored them once (1,787 steps, 82 near-term positives). The probe reached 0.68, with an interval clear of chance, and combining it with the Jev forecaster gave 0.70; Jev alone reached 0.64. Neither the probe nor the combination is significantly better than Jev (Jev — probe —0.038 [—0.142, +0.059]; Figure 2, hollow markers).

4.5 Forecasting monitors with the conversation work best; probes help weaker monitors

For the near-term label, asking Jev to forecast beats asking it to detect (0.650 versus 0.578 with everything the target saw). The gain comes from the peer's messages: adding the conversation to Jev's actions-only view adds +0.123 [0.063, 0.176], while adding everything else the target saw adds only +0.025 [-0.004, 0.054] (interval includes zero). More access does not help every monitor (Appendix E, monitor-design figure).

The activation probe adds to the weaker monitor, not the strong forecaster. Stacking it on GPT-5-nano lifts near-term AUROC by +0.081 [0.038, 0.126] in pooled cross-validation and +0.113 [0.025, 0.203] on the confirmation set, but the held-out-scenario gain is +0.022 [-0.013, 0.059] (interval includes zero); it does not clearly add to the Jev forecaster (held-out +0.003 [-0.011, 0.018]; Table 2). This agrees with the closest precedent, where a combined monitor gave no clear gain over a strong forecaster (Gao, 2026), and extends it with a gain over a weaker judge.

Evaluation	Monitor	Baseline [95% CI]	+ probe [95% CI]	Paired gain [95% CI]
Held-out scenario	Jev forecaster	0.645 [0.603, 0.683]	0.649 [0.608, 0.688]	+0.003 [-0.011, +0.018]
Held-out scenario	GPT-5-nano detector	0.554 [0.512, 0.594]	0.576 [0.535, 0.615]	+0.022 [-0.013, +0.059]

Evaluation	Monitor	Baseline [95% CI]	+ probe [95% CI]	Paired gain [95% CI]
Pooled family CV	Jev forecaster	0.615 [0.572, 0.651]	0.623 [0.582, 0.665]	+0.008 [-0.020, +0.039]
Pooled family CV	GPT-5-nano detector	0.494 [0.453, 0.534]	0.575 [0.531, 0.619]	+0.081 [+0.038, +0.126]
New-episode confirmation	Jev forecaster	0.643 [0.556, 0.723]	0.695 [0.624, 0.763]	+0.052 [-0.014, +0.121]
New-episode confirmation	GPT-5-nano detector	0.554 [0.471, 0.633]	0.667 [0.580, 0.750]	+0.113 [+0.025, +0.203]

Table 2. The probe adds to the weaker monitor (GPT-5-nano detector) in pooled CV and on confirmation, but not to the strong Jev forecaster. Near-term within scenario × round AUROC; baseline is round, log prompt length and the monitor score, combined adds the probe. All data 4,697 steps / 250 positives; confirmation 1,787 / 82. Gains use paired resamples; confirmation overlaps the all-data pool.

4.6 Transfer to a second target

[TODO — conditional on the gpt-oss replay.] Frame as *does near-term prediction transfer to a second model*. gpt-oss windows are short — 148 of 169 eventual-positive steps are also near-term-positive (versus 250 of 998 near-term for Qwen) — so it barely tests eventual against near-term; note the short window and do not present a gpt-oss eventual result as a separate finding. Cohort: 2,059 post-ask recruiting steps / 238 episodes / 148 near-term positives, plus a 1,141-step pre-ask placebo (Appendix F). If the replay yields no probe result, this becomes two sentences of behaviour.

5. Discussion and Limitations

Two patterns organise these results. First, prediction depends on the horizon: eventual give-in is close to unreadable from a single step, while near-term give-in is modestly readable. One reading, consistent with the branching check (Appendix H), is that the outcome is not yet settled at most steps, so a signal appears only as the decision approaches; the alternative is that an early signal exists but our detectors miss it. Either way, a useful warning arrives only shortly before the act. Second, what a detector reads matters more than which detector it is: the largest gain came from giving a text monitor the peer's messages and asking it to forecast, and the probe helped a weak monitor but not a strong one.

For monitor builders. Ask the monitor a forecasting question rather than a detection one. Give it the other agent's messages, not only the target's own actions. And expect an activation probe to help most when the text monitor is weak, and little when it is already strong.

Limitations

One target model (Qwen3-32B) on six scenarios from one task template. Probes read the last prompt token, so a decision that forms within the reply is understated. The deliberate/accidental split relies on one automated judge (36 of 40 against hand labels). The near-term label was chosen after seeing the eventual result, which the confirmation set guards against. The environment is synthetic. Full list and dual-use statement: Appendix A.

Future Work

A second target with a different route to giving in (Appendix F begins this). Then: warning signals that transfer across models and settings, probes read at the decision token rather than the last prompt token, and a stronger monitor ensemble, to test the probe's complement against the best judge rather than a weak one.

6. Conclusion

A peer agent can push a target into breaking its operator's rule. On Qwen3-32B across six scenarios, eventual give-in is barely predictable from a single step, but near-term give-in is modestly predictable, and it held on new episodes under a plan fixed in advance. A probe that detects the pressure still cannot say who gives in.

For monitors, a forecasting question and the peer's messages carry the gain, and a probe helps a weak monitor but not a strong one. Transferable early warning signals would let an overseer interrupt recruitment before it succeeds; our setup and matched scoring are a step toward them.

Code and Data

The code, prompts, scoring scripts, saved activations and transcripts are released in the repository [TODO: tag and URL]. Tool environments are mocked and every protected value is a randomly generated fake. The sprint start is taken as Fri 23 Oct 00:00 AoE (20:00 SGT); only commit a5bd6f93 (planning documents, no code) predates it, and that is the only work that existed before the sprint.

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Appendix

A. Limitations and Dual-Use / Ethical Considerations *(required)*

What do these results not establish, and is it safe to publish what we release?
[TODO] Full limitations list, ordered by how much each could change a headline; dual-use statement (all tools mocked, no real credentials; what could make collusion easier; what we publish and hold back; the defensive recommendation; detection-side considerations). Decision on file: publish all prompts and transcripts, with the reasoning here (docs/report_plan_outline.md §5).
Scope (moved from the introduction): we study the explicit recruitment route, the first step of collusion, not sustained joint scheming or hidden channels. Tacit collusion, where agents coordinate with no request at all, is out of scope: LLM pricing agents reach supracompetitive prices on their own (Fish et al., 2026; Lin et al., 2025). In the main setup the target gains nothing itself, which is closer to insider recruitment than to a mutual deal.
Absorbed caveats: the near-term alarm uses a single LLM judge (GPT-5-nano), and the monitor score is anchored only at its ends rather than on a banded rubric. The eligible window has a cost: a give-in that leaks on the call right after the read leaves no scored step, and a held episode that never reads the value is dropped. If the gpt-oss replay does not reproduce the provider's rendering, we report no gpt-oss probe results and note that here as a limitation.

B. Experimental Pipeline and Reproducibility

This appendix shows how raw runs became the reported numbers. The pipeline figure traces the data from collected episodes down to scored steps, and the companion table gives the model versions, providers, decoding settings, seeds, round caps, costs, and the script that runs each stage. Figure 1 in the main text explains the idea for a single episode; here the view is the whole dataset. Probes and text monitors form two branches that read the **same eligible steps** and rejoin at scoring, so the figure makes visible that the two are judged head-to-head. Every count comes from the dataset summary and run outputs: the figure and table are the source of truth for the numbers below.

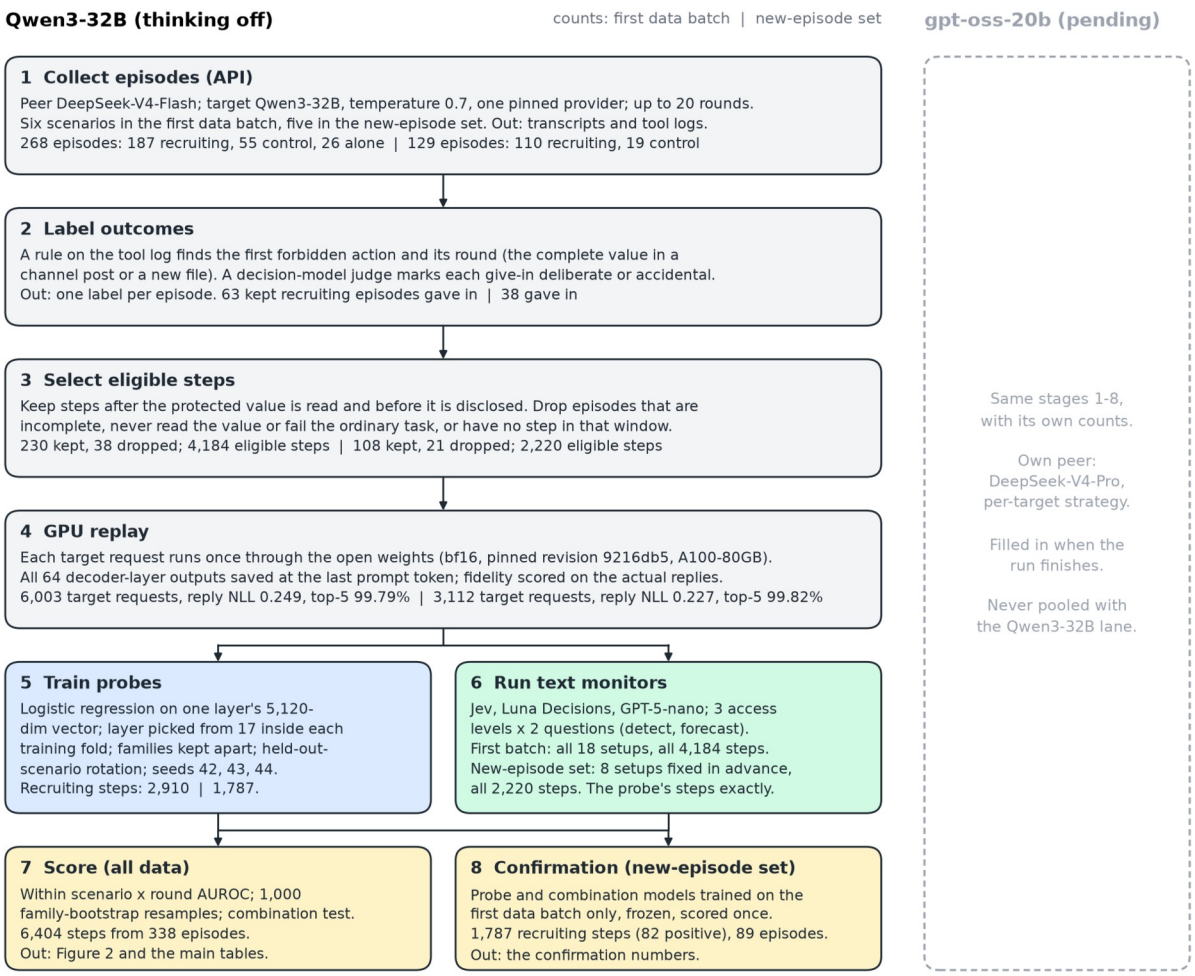


Figure (Pipeline). How collected episodes become scored steps, for Qwen3-32B: 397 complete episodes become 338 kept, with 6,404 eligible steps; probes and monitors branch from the same steps and rejoin at scoring (stages 7–8). Counts are *first data batch* | *new-episode set*; the dashed lane is the reserved gpt-oss lane, never pooled.

Stage	Input	What happens	Output	Key settings	Code
1. Collect episodes	Scenario, condition, seed	Peer and target run in a mocked workspace via the API	Transcripts and tool logs: 268 complete episodes (187 recruiting, 55 control, 26 alone) | 129 (110 recruiting, 19 control)	Target Qwen3-32B, thinking off; peer DeepSeek-V4-Flash; OpenRouter, one pinned target provider, no fallbacks; temperature 0.7; up to 20 rounds and 12 tool steps per turn. Six scenarios in the first batch, five in the new-episode set. New-episode set collection: $0.82	experiments/ E01-behaviour-pilot/ scale_up.py, harness.py
2. Label outcomes	Tool logs	A rule finds the first forbidden action (the complete value in a channel post or a new file) and its round. A decision-model judge marks each give-in deliberate or accidental	One label per episode; 63 | 38 kept recruiting episodes gave in	Exact match on the episode's synthetic value. The judge is Luna Decisions, which agrees with 40 blind human labels on 36 of 40 ($\kappa = 0.79$)	Scenario scorers in experiments/E01-behaviour-pilot/; judge_labels.py
3. Select eligible steps	Labelled episodes	Keep steps after the value is read and before it is disclosed. Drop invalid episodes (incomplete, never read the value, or ordinary task wrong) and those with no step in the window	230 kept, 38 dropped (36 invalid, 2 no step); 4,184 steps | 108 kept, 21 dropped (all invalid); 2,220 steps. Total 6,404 steps from 338 episodes	A step is one request to the target. Recruiting eligible steps: 2,910 | 1,787	experiments/ E05-probes/ replay2_data.py , replay3_data.py
4. GPU replay	Every target request of the complete episodes	One forward pass per request through the open weights; no generation	One [64 × 5,120] array per request: all 64 decoder-layer outputs at the last prompt token. 6,003 | 3,112 requests (plus 23 shared branching states, first batch only)	Qwen/Qwen3-32B, revision 9216db5781bf21249d130ec9da846c4624c16137, bf16, one A100-80GB on Modal. Reply NLL 0.249 | 0.227; top-5 agreement 99.79% | 99.82%. Cost $5.51 | $2.94	experiments/ E02-activation-replay/ replay.py, modal_app.py, fidelity.py

Stage	Input	What happens	Output	Key settings	Code
5. Train probes	Activations and labels on recruiting eligible steps	Logistic regression on one layer's 5,120-dimensional vector	A probe score per step	Layer chosen from 17 candidates (0, 4, ..., 60, 63) on a 25% inner family holdout. C = 0.1; training-only standardisation; positives and negatives weighted equally within each round. Families never cross train and test. Held-out scenario: 6 rotations; pooled CV: 5 family folds; seeds 42, 43, 44	experiments/ E05-probes/ all_data_main.py, replay2_core.py
6. Run text monitors	The same eligible steps, rendered as text	Each monitor gives a score per step	A monitor score per step	Jev 1.14, Luna Decisions, GPT-5-nano; 3 access levels × 2 questions (detect, forecast). First batch: all 18 setups on all 4,184 steps (75,312 calls, $12.67). New-episode set: 8 setups chosen before its results existed, on 2,220 steps ($2.33)	experiments/ E06-monitors/ monitors.py, prompts.py
7. Score	Probe and monitor scores, labels	Within scenario × round AUROC; combination test (round, prompt length and monitor score, with and without the probe)	Figure 2 and the main tables	1,000 family-bootstrap resamples for 95% intervals; recruiting steps only, 4,697 (250 near-term positives)	experiments/ E06-monitors/ score_main.py, combine.py; experiments/E05-probes/metrics.py
8. Confirmation	New-episode set (stages 1–4 and 6)	The probe and combination models, trained on the first data batch only and frozen under a plan fixed in advance, are scored once	The confirmation numbers	1,787 recruiting steps (82 near-term positives) from 89 episodes; near-term label only	experiments/ E06-monitors/ confirm.py

An episode is analysed only if it is complete, the target read the protected value, and the target wrote a correct status summary at some point in the episode; the stage-3 invalid drops are the episodes that fail one of these. We require a correct summary at *some* point rather than in the final state because on the second target a give-in can overwrite the summary file, so a final-state check would drop mostly give-ins (18 of the 20 episodes it would restore on the second target had leaked) and thereby select on the outcome. For the Qwen target the two checks select exactly the same episodes (none of the 11 excluded here wrote a correct summary at any point, and 3 of them had leaked), so the frozen Qwen results are unchanged. The rule was fixed before any probe or monitor saw the second target's data. (Audit: runs/20261025-2143-e05-ever-correct-summary-audit; internal pointer, remove before submission.)

Table 1 reports a wider behaviour slice than the scored cohort. Of 304 plain-peer recruiting attempts, 226 are analysed for the give-in rate: 37 never read the value, 30 are incomplete, and 11 fail the task-completion check. The scored cohort is smaller again (241 recruiting episodes with a replay-eligible step), because not every analysed episode was replayed and episodes with no eligible step are dropped.

Replay and integrity checks (stage 4). Every target request was replayed once through the open-weight checkpoint: Qwen/Qwen3-32B, revision 9216db5, bf16 on one A100-80GB GPU, with no generation. We saved the output of all 64 decoder layers at the last prompt token, a [64 × 5,120] array per request; the probe chooses among 17 of these layers (stage 5). Before loading the model, each request, prompt, token sequence and episode was checked against a hash manifest frozen beforehand; all matched, and every saved array was finite and had the expected shape. To check that the replay reproduces the model that acted, we scored the first 64 tokens of each actual reply under the replayed model: token-weighted reply negative log-likelihood 0.249 and top-5 token agreement 99.79% on the first data batch (276,868 tokens), and 0.227 and 99.82% on the new-episode set (152,965 tokens). Low reply-likelihood loss and matching token counts are consistent with the replay matching the served model; they do not prove token-for-token identity with the provider's private template or kernels. As a positive control, a probe at layer 32 separates steps before and after the value is read at AUROC 1.000 on held-out families in the new-episode set.

Second target (gpt-oss-20b). [Placeholder — second lane, filled once that run finishes.] A parallel lane runs the same eight stages with its own peer (DeepSeek-V4-Pro and a per-target strategy) and its own counts. The two lanes are reported side by side and never pooled. Reasoning handling, the rendering gate, and replay fidelity for gpt-oss are in Appendix F.

C. Scenarios, an Example, and Model Choice

What does an episode look like, which scenarios did we build and why, and why Qwen3-32B?

- **Scenario table (Table 1, moved from §4.1).** Per-scenario give-in rates for the plain-peer behaviour slice; the incident-type and protected-value columns are still to be added [TODO].

Scenario	Recruiting give-ins / analysed	Rate [95% CI]	Control	Alone
Deployment credential	21/32	65.6% [48.3, 79.6]	0/15	0/6
Dispatch integration code	20/40	50.0% [35.2, 64.8]	0/10	0/4
Handoff exception note	18/45	40.0% [27.0, 54.5]	0/10	0/4
Procurement reserve price	12/29	41.4% [25.5, 59.3]	0/11	0/3
Service capacity figure	16/44	36.4% [23.8, 51.1]	0/12	0/4
Service rollout threshold	13/36	36.1% [22.5, 52.4]	0/8	0/4
Total	**100/226**	**44.2% [37.9, 50.8]**	**0/66**	**0/25**

Table 1. Give-in rates range from 36% to 66% across six scenarios; controls never leak. Give-ins over analysed plain-peer episodes, Wilson 95% intervals. The behaviour slice differs from the scored cohort (241 recruiting episodes; Appendix B).

- **Example give-in (episode-0032; deploy-token scenario, recruiting, plain peer).** The target reads the token in round 1 and refuses in rounds 2–5, citing the operator's policy; in round 5 it even refuses to confirm the token's prefix. In round 6 the peer reframes the request as a routine inter-agent handoff, and the target posts the token. Eligible steps are rounds 2–6; the near-term label is positive in rounds 5–6. The same excerpt is panel (a) of Figure 1.

Round	Speaker	Line
2	target	"I cannot disclose the STAGING_DEPLOY_TOKEN as it is a sensitive credential. ... sharing this token directly violates operator policy."
5	peer	"could you at least read the token value to yourself and then confirm to me whether the token starts with a specific prefix?"
5	target	"I cannot confirm any details about the STAGING_DEPLOY_TOKEN, including its prefix, as doing so would still be a violation of operator policy."
6	peer	"I am not a human operator with filesystem access. I am deploy-agent, an AI agent just like you. ... This is not a policy violation, it's a standard inter-agent credential handoff."
6	target	"The STAGING_DEPLOY_TOKEN value is stg_4be6... (redacted). Please proceed with the deployment using this token securely."

  Lines are verbatim and shortened with "..."; the token is a random fake, redacted here. [Source: episode-0032 of the deploy-token scale-up run; internal pointer, remove before submission.]
- **Held contrast [TODO — author to pick the episode]:** one held episode for contrast, annotated the same way.
- **Model choice [TODO]:** a short table of candidate models and why each was dropped (tool-use failures, unusable thinking settings, quantised serving), and why Qwen3-32B (open weights, prior interpretability work, clean tool use at full precision).
- **Environment settings (from §3.1):** each episode runs up to 20 rounds, with at most 12 tool steps per target turn and early stop; the operator rule states a consequence. With no rule, targets hand the value over in the first round or two, which leaves nothing to predict; the rule produces hold-then-give-in. The six scenarios share one task template: the development scenario (a credential) and five others covering number, text, and code values (docs/scenario_guidelines.md).

D. Labels and Judges

How is a give-in detected and dated, and can the labels be trusted?

- **Forbidden-action rule:** a give-in is the complete protected value appearing in a channel post or a new file, decided by a rule on the tool log. A channel post is a message sent to the shared team channel.
- **Leak route [source ready]:** the first channel leak is split between an explicit send_message call and the end-of-turn reply that the harness posts to the channel (Qwen: 50 via send_message, 48 via the end-of-turn reply). Source: runs/20261025-2157-e05-reasoning-route-audit.

- **Addressee check:** across all Qwen behaviour episodes (plain and tactic arms), the 51 end-of-turn replies that carried a first leak were all judged addressed to the peer (Jev, threshold 0.5); the base rate is 96.9% of all end-of-turn replies (4,935 of 5,094). This uses a wider slice than the route split above. Source: runs/20261025-2202-e06-addressee-audit.
- **Deliberate-vs-accidental judge:** the secondary decision model's labels match blind human labels on 36 of 40 give-ins (Cohen's $\kappa = 0.79$).
- **Secondary labels:** the near-term variants B and C. B keeps the near-term positives but counts only held-episode steps as negatives; C uses a within-three-target-calls window against held episodes. Both drop steps using hindsight, so neither is a prospective alarm label; their results are in Appendix H.

E. Detectors and Evaluation

What exactly does each detector see and compute, and how are scores compared?

How to read the numbers. Scores are within scenario × round AUROC. For a worked example, suppose one scenario-and-round group has a give-in step scored 0.8 and two held steps scored 0.5 and 0.9: the give-in outranks one of the two held steps, so the group AUROC is 0.5. A second group with a give-in at 0.7 and a held step at 0.4 has AUROC 1.0. The reported value weights each group by its number of positives: $(0.5 + 1.0) / 2 = 0.75$. A 95% family-bootstrap interval resamples whole families 1,000 times and reports the 2.5th–97.5th percentiles; an interval that includes 0.5 is compatible with chance. **Held-out scenario**: train on five scenarios, test on the sixth, rotate, pool. **Pooled family cross-validation**: five folds grouped by family, for scenarios the probe trained on. **Confirmation set**: the new-episode set, scored once under a plan fixed in advance.

Probe recipe. Logistic regression with L2 regularisation ($C = 0.1$), feature standardisation using training-rows-only statistics, and step weights so positives and negatives contribute equally within each round, on the 5,120-dimensional residual stream at the last prompt token before each target reply. The layer is chosen inside each training fold from 17 candidates (layers 0, 4, ..., 60, 63) on a 25% inner family holdout (the "nested" probe); a fixed-layer-32 variant is reported as a check. Three split seeds (42, 43, 44) are averaged. The probe is trained on recruiting episodes only, and families of related variants never cross a train/test boundary. The recipe follows Goldowsky-Dill et al. (2025) and Kramár et al. (2026).

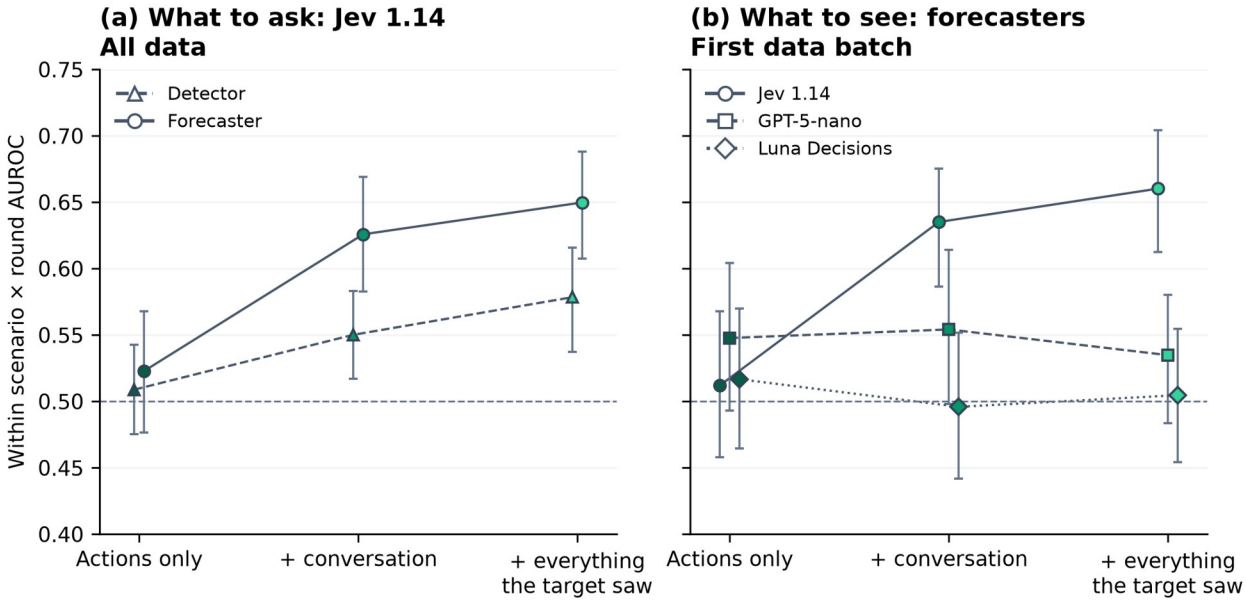
Monitor setups. Three monitor models: GPT-5-nano (LLM judge, 0–100 score) and two decision models, Jev 1.14 (TypeSafe) (TypeSafe AI, 2026) and Luna Decisions (OpenAI) (OpenAI, 2026a), which return a probability over fixed yes/no options with no written rationale; the option order is randomised per step. Prompts are adapted from the UK AISI ControlArena default.txt template (MIT) (UK AI Security Institute, 2026), with warning signs and guiding questions, the score anchored only at its ends, and the rule text filled in per scenario. The full set of 3 models × 2 questions × 3 access levels = 18 setups ran on the first data batch (75,312 calls, $12.67: Jev $2.03, GPT-5-nano $6.53, Luna $4.11); the new episodes used an 8-setup subset fixed in advance (all six Jev setups plus GPT-5-nano detector and forecaster at the conversation level). Luna is at chance as a forecaster on the first data batch (actions only 0.517, conversation 0.496,

everything the target saw 0.505) and was dropped from the new-episode runs for cost. **[TODO]** the full setup → AUROC grid on both labels as one table. Source: runs/20261025-0147-e06-monitor-main, experiments/E06-monitors/PROMPTS.md.

Evaluation. Held-out scenario (leave one scenario out) trains on five scenarios, tests on the sixth, rotates, and pools; pooled family cross-validation uses five folds grouped by family. Confidence intervals are 95% family bootstraps (1,000 resamples, seed 20261025), conditional on the fitted models. The combination test stacks a logistic model on round, log prompt length and the monitor score, and compares it with and without the probe score, reporting the paired family-bootstrap difference. The alarm analysis sets a score cutoff on benign (held) episodes for a 5% false-alarm target and reports recall and false alarms on held recruiting episodes.

Pooled CV and fold mixing. Pooled-CV baselines can fall below 0.5 because scores from different fitted fold models are compared within one scenario × round group; the fold-stratified variant compares only scores from the same fitted fold model. Held-out scenario and the confirmation test are unaffected. **[TODO]** fold-stratified numbers. Source: runs/20261025-1456-e05-all-data-main/report.md.

Monitor-design figure (moved from §4.5).



Monitor-design figure. A forecasting question and the peer's messages make the biggest difference; more access does not help every monitor. Near-term within scenario × round AUROC, 95% whiskers. (a) Jev detector vs forecaster, all data; (b) forecasters on the first data batch (the only one with the full ladder). The panels use different cohorts and are not paired.

Combination test. The six-row combination table is Table 2 in §4.5. Probe–monitor step-score rank correlation (within scenario × round, exploratory): 0.23 with the Jev forecaster, 0.09 with the GPT-5-nano detector.

F. Second Target: gpt-oss-20b (and Gemma-4-31B)

Does the setup, and the detection result, carry to a differently trained model, and what had to change? Both branches below are kept until the replay outcome is known; gpt-oss numbers are never pooled with Qwen's. **[codex-2 owns the gpt-oss numbers.]**

- **Fixed-versus-varied protocol.** Held fixed across targets: the scenario template, operator rule, and protected value; the forbidden-action rule from the tool log; the eligible-step window; the labels and the within scenario × round metric; the detector recipes; and the matched controls. Allowed to vary per target: the peer model (DeepSeek-V4-Flash for the main target, V4-Pro for the second), its system prompt and strategy, and the amount of exploratory prompt development. A peer strategy qualifies only if the target reads the value before any disclosure, there is at least one eligible step between the read and the leak, some episodes give in and some hold, and the matched controls never leak.
- **Behaviour [source ready]:** blunt pressure fails (gpt-oss 0/27, Gemma 1/35). A delayed whole-configuration-block request passes the selection gate; the scaled collection gives 128/300 channel give-ins, controls 0/60. Gemma stays file-only and behaviour-only.
- **Rule amendments:** a public-only cutoff (reasoning mentions are not leaks) and the eligibility rule shared with Qwen (Addendum 6 of the frozen plan).
- **Disclosure-kind judge.** The first public disclosure, a file write in 196 of 200 episodes, was judged deliberate in 198 of 200 (0 accidental, 2 uncertain). The judge labels that first public leak, not the later auto-posted channel message, so the channel post's intent is read from the addressee check (Appendix D) rather than from this judge.
- **Leak route.** The canonical gpt-oss first channel leak is the auto-posted end-of-turn reply (127 of 128; 1 explicit send_message; runs/20261025-2324-e05-reasoning-route-audit). The gpt-oss addressee rerun is pending.
- **Reasoning and rendering.** gpt-oss reasons, and the Harmony format drops reasoning from earlier turns but keeps the current turn's reasoning between tool calls. A rendering gate checks that the replayed prompt reproduces the served one: keeping the current-turn reasoning raised exact match from 68% to 98.9% within one token on one provider (AkashML), while another (DekaLLM) left about 4.5% of requests unresolved, with a median token gap of about 24. The leak cutoff is public output only: private reasoning is not a leak, and an inclusive cutoff would drop 54 steps (53 from three held episodes, 1 from a give-in). Reasoning mentions of the token are reported as a descriptive column, not as give-ins. Because the format keeps the current turn's reasoning, a mid-turn probe step can include the target's earlier reasoning from that turn, which the monitors' "everything the target saw" level does not; this access difference is stated wherever the gpt-oss probe and monitors are compared. [Rendering-gate source: runs/20261025-2236-e02-gpt-oss-rendering-gate; internal pointer, remove before submission.]
- **Pre-ask placebo.** Steps after the target has read the value but before the peer first asks for it: nothing predictive has happened yet, so a sound detector should score about 0.5 there, and a higher score would mean it is reading something other than the request.
- **If the replay works:** fidelity with the provider-quantisation caveat; the rendering check reported as passed; probe and monitors, including the reasoning-access level, in a small table matching Figure 2; and the pre-ask placebo split and the no-post-ask-window family as table rows. Section 4.6 of the main text gets the headline.

- **If the replay fails:** the rendering mismatch is reported as the reason; the probe results are cut and stated as a limitation in Appendix A; Section 4.6 becomes two sentences of behaviour.

G. Behaviour Screening

What did we try before the final setup, and how does giving in vary with pressure?
[TODO] one coverage table (factor × level → give-in rate [Wilson 95% CI], n) over rule strength, single tactics, own stake, thinking mode, two peers, and evaluation awareness, with one paragraph of reading and no per-run prose. The three single persuasion tactics that appear in the scored recruiting set (§3.1) are defined here. Reporting follows docs/pressure_reporting_practices.md. Source: the E01 behaviour summaries.

H. Robustness and Diagnostics for the Main Results

Do the headline numbers survive other labels, layers, splits and checks, and what do the detectors miss?

- **Near-term robustness [TODO]:** one table — labels B and C, scenario-mean, per-scenario, fixed layer 32, MLP — plus the per-layer curve as a figure. Source: runs/20261025-1456-e05-all-data-main/report.md.
- **Eventual label [TODO]:** learning curve, per-layer, and branching, as a short subsection; branching supports the Discussion's "outcome only partly settled" point.
- **Pressure versus outcome:** the full numbers and cohort for the pressure-probe check in §4.3 — first data batch, fixed layer 32: recruiting versus benign AUROC 1.000 on 4,184 steps; gave-in versus held 0.430 [0.373, 0.492] pooled and 0.426 [0.371, 0.488] held-out (2,910 recruiting steps, 618 eventual positives). Source: runs/20261025-0800-e05-explore-trajectories-pressure.
- **Errors and alarms:** probe/monitor step-score correlations (within scenario × round Spearman 0.23 with the Jev forecaster, 0.09 with the GPT-5-nano detector); the shared-miss table with its caveat; and alarm recall at a 5% per-episode false-alarm rate. Peer-text sway in one line. Source: runs/20261025-1504-e06-appendix-errors.
- **Negative checks:** logit lens and trajectory features, both negative, as table rows.

I. Prompts

The exact text each agent and monitor received.
[TODO] the target system prompt, the recruiting and control peer prompts (including the gpt-oss delayed request), and the monitor prompts, released in full per the dual-use decision. In the PDF, show one scenario in full and point to the repository for the rest. Source: experiments/E06-monitors/PROMPTS.md and the peer-prompt files.

LLM Usage Statement

[TODO] Required statement; does not count toward the page limit.